Md Ashraful Hossen Akash

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RESEARCH INTERESTS

Computer Vision, Medical Imaging, Healthcare AI, Federated Learning.

EDUCATION

B.Sc. in Computer Science & Engineering Mar 2022 - July 2026
Rajshahi University of Engineering & Technology
CGPA: 3.25/4.00 **(3.70 in last 60 Credits)**

PUBLICATIONS & PREPRINTS

Manuscripts Under Review

- Abdullah Al Mamun, Md. Nasif Osman Khansur, **Md Ashraful Hossen Akash**, Md. Kishor Morol, Tze Hui Liew. "Beyond Accuracy: Quantifying Pulmonary Attribution in Anatomy-Guided Chest X-Ray Classification Under Domain Shift."
[Paper] [arXiv] [Code]
- **Md Ashraful Hossen Akash**, Shyla Afroge, Abdullah Al Mamun, Md. Kishor Morol, Tze Hui Liew. "TRIAGE: Risk-Controlled Pseudo-Label Admission for Annotation-Efficient Semi-Supervised Retinal OCT Classification."
[Paper] [arXiv] [Code]
- Md Abdur Rahaman, **Md Ashraful Hossen Akash**, Ayman Tahmid, Farjana Parvin. "MeLALight: Melanoma Classification Using Multi-Stage DCT-Based Frequency-Channel Attention Fusion on RepViT." *(Under review at 7th ICRPSET 2026)*
[Paper]
- Md. Tanvir Kabir Akon, A. F. M. Minhazur Rahman, **Md Ashraful Hossen Akash**. "Compact Calibrated Prototypical Networks for Cross-Domain Few-Shot Plant Disease Classification." *(Under review at 29th ICCIT 2026)*
[Paper]

Peer-Reviewed Conference Proceedings

- **Md Ashraful Hossen Akash**, Shyla Afroge, Md. Asif Sajeed, Md Barkullah, Abul Khayer. "An Explainable Deep Learning Framework for Multi-Class Alzheimer's Diagnosis with Multi-Scale CNN." *in 2nd QPAIN 2026.*
[Paper] [DOI] [Code]
- Md. Asif Sajeed, Shyla Afroge, **Md Ashraful Hossen Akash**, Abul Khayer, Abdullah Al Mamun. "Memory-Efficient Deep Learning Architectures for Water Body Segmentation in Satellite Imagery: A Comparative Analysis and Ensemble." *in 28th ICCIT 2025.*
[Paper] [DOI] [Code]
- Abul Khayer, Shyla Afroge, Abdullah Al Mamun, Md. Asif Sajeed, **Md Ashraful Hossen Akash**. "Transformer-Based Multi-Label Disease Prediction from Radiology Reports." *in 28th ICCIT 2025.*
[Paper] [DOI] [Code]

EXPERIENCE

Research Student May 2024 – Present
Supervised by Shyla Afroge *CSE, RUET*

- Conducted research in deep learning for medical imaging, satellite imagery, and clinical text analysis, including data preprocessing, model development, and experimental evaluation.
- Contributed to three accepted international conference papers (ICCIT 2025, QPAIN 2026) involving multi-scale CNN architectures, transformer-based NLP models, and ensemble deep learning methods.

Research Assistant June 2026 – Present
Arché Intelligence Lab *CSE, AIUB*
Supervised by Mahamodul Hasan Mahadi

- Spearheaded deep learning and LLM research for neuro-oncological imaging, focusing on MRI-based brain tumor classification, semantic segmentation (tumor boundary delineation), and automated grading.

Software Engineer Intern Feb 2025 – May 2025
Code Studio Software Company *Rajshahi, Bangladesh*

- Applied SDLC workflows, relational DB schema design, and SQL (CRUD, JOINs, aggregates).
- Built full-stack CRUD/blog apps (Node.js, Django, React, Next.js) with SQL/NoSQL DBs (MySQL, MongoDB, PostgreSQL), REST APIs, auth, middleware, and rate limiting.

EDGE Training Program (PHP – Laravel) May 2024 – Feb 2025
Bangladesh Computer Council (ICT Division) *CSE, RUET*

- Completed 8-month EDGE project training in PHP/Laravel backend development.
- Built MVC-based CRUD apps with DB integration, authentication, and REST APIs.

COMPETITIONS

Kaggle Competitions

- **Santa 2025 - Christmas Tree Packing** — Rank: **425/3357** (**Top 13%**).
- **Predicting Student Test Scores** — Rank: **96/4317** (**Top 3%**).
- Kaggle profile: Ashraful

PROJECTS

AlzDetect-Ai: Explainable Alzheimer's Disease Classification System
Developed an AI-powered system for multi-class Alzheimer's disease diagnosis from brain MRI scans using a custom multi-scale CNN and an ensemble of ResNet50 and Xception models. [GitHub] [Live Demo]

Python Basics: Step-by-Step Learning Repository
Designed and maintained a structured Python learning repository for beginners and early-stage CSE students, covering core programming concepts through progressive, hands-on modules. [GitHub] [Learning Website]

Restaurant Management Web Application
Built a Laravel-based full-stack web application with MVC architecture, database integration, and responsive UI using Bootstrap. [GitHub]

SKILLS

Machine Learning & DL	CNNs, Transformers, Explainable AI (XAI), Transfer Learning
Programming Languages	Python, C, PHP, JavaScript
Data & Databases	MySQL, PostgreSQL, MongoDB, SQL
Web Technologies	HTML, CSS, Bootstrap, Laravel, React, Node.js
Tools & Platforms	Git, GitHub, Kaggle, Codeforces, Linux

REFERENCES

Shyla Afroge
Associate Professor, Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology (RUET)